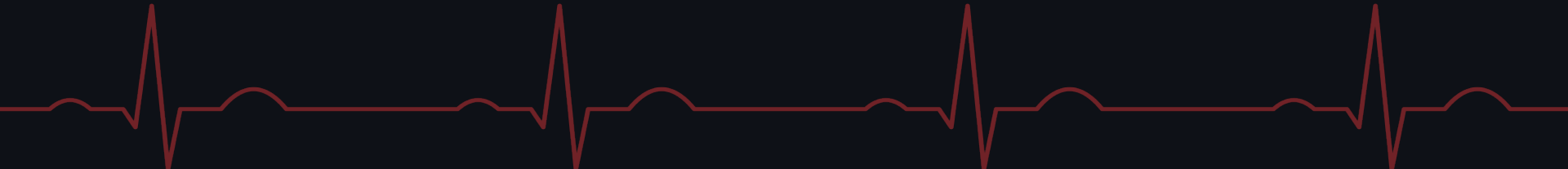


NATURE

*enVast-assisted mechanical thrombectomy versus standard primary PCI
in STEMI with large thrombus burden*



154

PATIENTS

11

CENTRES

1:1

RANDOMISED

-26%

INFARCT SIZE

Large thrombus: the clot that won't stay put

1

Large thrombus burden

Common in STEMI; ballooning and stenting can fragment the clot.

2

Distal embolisation

Debris showers into the microcirculation downstream of the culprit lesion.

3

Microvascular obstruction

Epicardial flow is restored, but tissue-level reperfusion fails.

4

Larger infarct

More myocardium lost, higher risk of heart failure and death.

Aspiration thrombectomy already failed



TASTE

n = 7,244 · no mortality benefit from routine aspiration



TOTAL

n = 10,732 · no clinical benefit; excess stroke at 30 days



Guidelines

Routine thrombus aspiration not recommended (ESC Class III)

enVast: a stent-retriever built for coronary clot

Developed by Vesalio from its NeVa neurovascular thrombectomy platform, enVast captures and removes thrombus mechanically before the operator balloons or stents the culprit lesion.



STEP 1

Cross & deploy

Wire the infarct-related artery and deploy the retriever across the thrombus.



STEP 2

Capture

The clot is engaged and held within the device's struts.



STEP 3

Retrieve, then PCI

Withdraw device and clot together, then complete conventional PCI.

Prospective, multicentre, randomised 1 : 1



154 STEMI patients

large thrombus burden · 11 centres



RANDOMISED 1 : 1



enVast thrombectomy
then conventional PCI



Standard of care
conventional primary PCI

Follow-up: CK-MB AUC · CMR at day 3 · 30-day clinical events

HYPOTHESIS

Superiority of enVast on infarct size

PRINCIPAL INVESTIGATOR

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Ticino, Lugano

SPONSOR

Vesalio

REGISTRY

NCT04969471



Inclusion and exclusion criteria



Inclusion

5 criteria

- Age ≥ 18 years
- Chest pain > 20 min with ST elevation ≥ 1 mm in ≥ 2 contiguous leads, or infero-lateral MI with ST depression ≥ 1 mm in ≥ 2 of V1–V3 and a positive terminal T wave
- TIMI thrombus grade ≥ 3 in the infarct-related artery (re-confirmed after wiring if TIMI 0 flow)
- Intervention started within 8 h of symptom onset
- Informed consent before the procedure



Exclusion

7 criteria

- Unconscious patient
- Infarct-related artery < 2.5 mm (visual estimate)
- Severe calcification or extreme tortuosity at or proximal to the culprit lesion
- Stent thrombosis as the culprit lesion
- Previous MI in the same territory
- Women of child-bearing potential
- Participation in another interventional trial

ENDPOINTS

Enriched for clot; judged on infarct size

ENRICHED FOR CLOT

TTG ≥ 3

TIMI thrombus grade 3 or more: a clot longer than half the vessel diameter, in the patients most at risk of distal embolisation.

Why enrich? Benefit is most plausible where there is the most clot to embolise.



PRIMARY EFFICACY

Infarct size by CK-MB area under the curve



KEY SECONDARY

Infarct size on cardiac MRI (% LV mass) at day 3



SAFETY

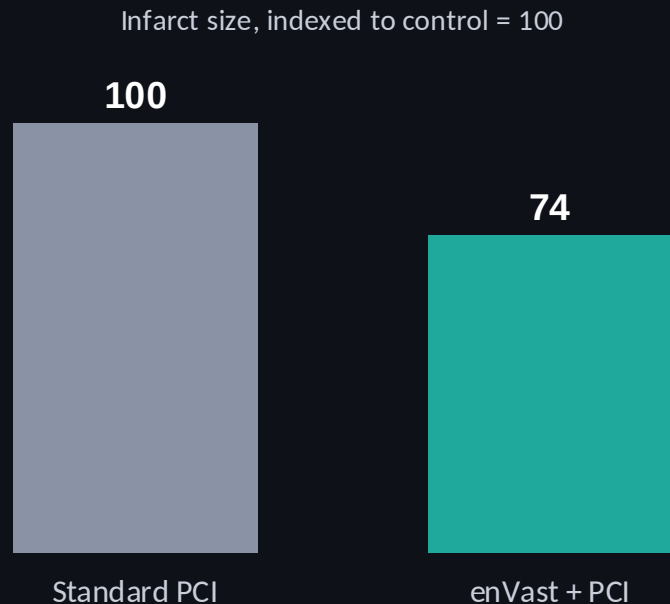
Death, stroke and MACE at 30 days

Infarct size cut by about a quarter

-26%

relative reduction in infarct size (CK-MB AUC) with
enVast versus standard PCI

P = 0.001



Biomarker and MRI tell the same story

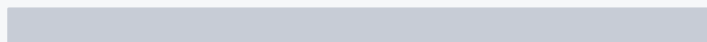


CK-MB AUC

Primary endpoint

-26%

Standard PCI



enVast + PCI



P = 0.001 • significant

≈

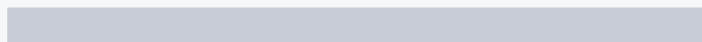


Cardiac MRI, day 3

Infarct size, % LV mass

-25%

Standard PCI

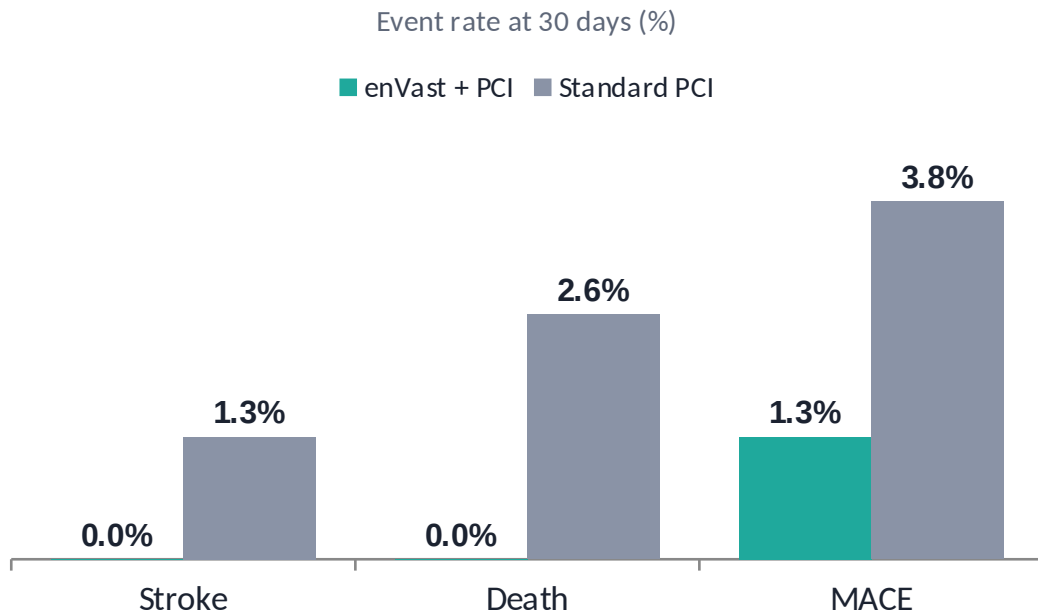


enVast + PCI



Directionally consistent trend

No strokes, no deaths in the enVast arm



Why it matters

Aspiration in TOTAL carried a stroke signal. NATURE saw none with enVast.



Read with care

Only a handful of events; the trial was not powered for clinical outcomes.

Strong signal, early evidence



Strengths

- Randomised, multicentre design
- Population enriched for large thrombus — where benefit is most plausible
- Biomarker and CMR results agree
- No stroke or death signal at 30 days



Limitations

- Modest sample (n = 154)
- Surrogate primary endpoint, not hard outcomes
- Operators cannot be blinded to device use
- CMR effect a trend; industry-sponsored

VERDICT

Confirms the hypothesis on infarct size; generates the hypothesis for clinical outcomes.

Three things NATURE showed

01

Superior on infarct size

enVast before PCI cut enzymatic infarct size by ~26% in large-thrombus STEMI ($P = 0.001$).

02

Consistent and safe

CMR pointed the same way; no strokes or deaths with enVast at 30 days.

03

Right patient, right time

Benefit shown where it is most plausible: a large thrombus, treated within 8 hours of symptom onset.

CONCLUSION

“ In STEMI with a large thrombus burden, enVast-assisted thrombectomy before PCI was **superior to standard PCI** in reducing infarct size, with no early safety penalty.



For practice today

A promising option for selected large-thrombus cases, not yet a routine strategy. Guidelines stay unchanged until outcome data arrive.



For research next

An adequately powered randomised trial with hard clinical endpoints: death, heart failure and stroke.



References



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